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Verification of a porous media model for the partial oxidation of a chemically reacting fixed-bed

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ABSTRACT

In this work, two numerical approaches, namely particle-resolved simulations (3D PRS) and porous media model (PMM), are assessed using a 2D axisymmetric CFD pseudo-state approach (PSS) for validation of the gasification of carbon char particles in a chemically reacting fixed-bed. Simulations are performed using the commercial software Ansys® Fluent, implementing a six-reaction semi-global scheme, including three heterogeneous, and three homogeneous chemical reactions. The effects of the inflow gas temperature ($T_{in} = 900\text{--}1500\text{ K}$), mass fraction of oxygen ($Y_{O_2,in} = 0.05, 0.11$), and laminar flow regime ($Re_{in} = 50, 75$ and 100) reveals a significant role of T_{in} , observing important deviations at $T_{in} = 1300\text{ K}$. Minimum deviations when comparing 3D PRS and 2D PMM models are observed for $Y_{O_2,in} = 0.05$, being 3.6% for CO, 5.5% for CO₂ and 0.76% for temperature inside the porous zone at different operating conditions. Thermal conductivity of the solid material, axial dispersion coefficient, and radial variation of porosity are analyzed and discussed, showing good agreement between numerical models, remarking the role of geometrical features of the unit in the gasification process.

1. Introduction

Filtration combustion (FC) is the process where an oxidant stream flows through a porous media reactor composed of a solid inert/fuel material, involving exothermic chemical transformations [1], for example, in the gasification of carbonaceous materials in fixed-bed configuration [2]. These units have been widely studied since their multiphase and multiscale modeling for applications such as steam methane reforming (SMR) for hydrogen production and energy storage [3,4], catalytic combustion [5] and pyrolysis [6], among others, are key for the chemical industry, and specially challenging from a numerical perspective [7]. In particular, for the thermochemical conversion of solid fuels, the thermogravimetric analysis (TGA) has shown similar kinetic modeling of thermochemical processes [8], with applications of coal char gasification [9] and sawdust and coal blend pyrolysis [10]. The heterogeneous interaction between different phases, most likely solid and gas, leads to a transient approach to the excess of enthalpy theory [11]. The theory is based on the intensified radiation heat

transfer mechanism when the inflow gas mixture meets the combustion zone, taking advantage of the high radiant emissivity of the solid fuel. Thus, the combustion temperature may exceed the theoretical adiabatic temperature rise since there is a preheating zone in the inlet section of the unit. There are several experimental studies in this field that have used different fuel/oxidant mixtures, such as methane, air, and oxygen, among others [12]. Filtration combustion may be extended to inert porous media combustion when there are non-chemically reacting particles inside of the reactor [13], e.g., alumina spheres, or to hybrid filtration combustion (HFC), when fuel and inert solid particles coexist inside the bed. Under this configuration, that might be described as the combination of inert porous media combustion and chemically reacting fixed-bed phenomena, both heterogeneous and homogeneous chemical reactions achieve high temperatures in the range of 900–2300 K, observing better performance for H₂ and syngas at temperatures up to 1800 K [14]. Salgansky et al. [15] studied the gasification of a solid fuel using a vapor–air mixture as an oxidant agent in a filtration mode, and

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Nomenclature	
Roman Symbols	Definition
A	Pre-exponential factor
C	Concentration, kmol m ⁻³
D	Mass diffusion coefficient, m ² s ⁻¹
d	Diameter, m
E_a	Activation energy, J mol ⁻¹
h	Enthalpy, J kg ⁻¹
k	Reaction rate constant
L	Length, m
M	Molecular weight, kg mol ⁻¹
$\dot{m}''$	Net mass flux gas-surface, kg m ⁻² s ⁻¹
N	Number
p	Pressure, Pa
R	Universal gas constant, J mol ⁻¹ K ⁻¹
$\hat{R}$	Production rate, mol m ⁻² s ⁻¹
Re	Reynolds number
S'''	Specific surface area, m ⁻¹
T	Temperature, K
$\vec{u}$	Velocity, m s ⁻¹
Y	Mass fraction
Greek Symbols	Definition
ϵ	Emissivity
ε	Porosity
λ	Thermal conductivity, W m ⁻¹ K ⁻¹
μ	Viscosity, Pa s
ρ	Density, kg m ⁻³
τ	Stress tensor, Pa
Abbreviations	Definition
CFD	Computational fluid dynamics
FC	Filtration combustion
PMM	Porous media model
PRS	Particle-resolved simulation
Subscripts	Definition
C	Carbon char particle
i	Species i
in	Inlet
p	Particle
pz	Porous zone
R	Reaction
t	Tube

assessed the length of the reactor and the relevance of thermodynamics in the mathematical calculations for gasification efficiency. Later, a superadiabatic regime due to filtration combustion was evaluated for a carbon-inert bed. Agreement was observed between theory and experiments by assessing the temperature of the filtration wave, composition of gaseous products, and yield of carbon monoxide and hydrogen [16].

Several studies have been carried out on the oxidation of carbon from a numerical perspective using a single spherical particle. In particular, studies have explored the steady-state combustion of a coal particle [17], the oxidation of a single moving particle under laminar regime conditions [18], and the consumption rate of carbon based on the effect of critical parameters such as inlet velocity, composition, and temperature [19]. using a detailed model in which surface oxidation and volatilization of the coal particle were included. It has been found in the literature a strong relationship between the particle Reynolds number (Re_{in}) and oxidation regimes related to species transport to the particle's surface. Nevertheless, for a chemically reacting porous medium, and to the best of the author's knowledge, research regarding the applicability of a porous media model and its validation against a three-dimensional CFD particle resolved numerical approach is scarce. On the other hand, the heterogeneous combustion phenomenon has

been identified as a challenge specifically to char since an important amount of physical processes are taking place on the surface of the particles, being widely studied for approximately a hundred years now [20]. Furthermore, numerical simulations for studying carbon thermochemical conversion have been reported in the literature in the past decades [21–26]. Libby and Blake [21] studied the effect of water vapor on the carbon consumption in a hot ambient, while Lee et al. [22] and Cho et al. [23] explored the interactions between heterogeneous chemical reactions, mass and energy transport in and around chemically reacting particles. Interesting findings relate to the relationship between burning rate and oxygen concentration at the inlet of the reactor [24]. Furthermore, for the thermochemical conversion at low concentration of oxygen in a fixed-bed reactor, radiation has been identified as the dominant heat transfer mechanism, highlighting the relevance of heterogeneous reactions and the phase change of water along the process [25,26]. Most recently, a hybrid particle model has been developed to assess the char conversion process, considering the effect of pore diffusion on the carbon consumption rate inside gasifiers [27]. Moreover, carbon char thermochemical conversion has been studied in a fixed-bed gasifier using surface-based reactions [28] using a 3D particle-resolved simulation (PRS) approach, which will be the baseline of future analysis in this present article.

Separately, it should be noted that a comparative study between the so-called 3D particle resolved (PR) CFD-based model and the porous media based model has been presented by Dixon [4]. The study involved conducting 3D CFD-based PR simulation, comparing local heat and mass transport characteristics, and assessing reaction rates in a fixed-bed reactor tube referring to endothermic steam methane reforming. Vardhan Reddy Kuncharam & Dixon [29] presented a new multi-scale non-isothermal heterogeneous steady-state model for steam methane reforming in a packed-bed tube. The model is based on a porous media approach where two-dimensional transport equations describing the heat and mass transfer in the gas phase are directly coupled with the chemical reactions (including the heat and mass transfer) inside the catalyst particle. The results of a sensitivity analysis studying different closure relations (empirical equations) were discussed. The simulations were conducted on a 10 m long packed tube.

The novelty of this work consists of performing numerical studies of the intraparticle heat and mass transfer in a chemically reacting fixed-bed, where numerical models are based on the so-called particle-resolved (3D PRS) and porous media (2D PMM) approach CFD simulations. The use of a PMM approach, which is computationally suitable for modeling a large number of porous particles (200–300 particles normally used in experiments), is necessary to validate and could potentially enable the use of the PMM approach in particle-resolved fixed-bed units with porous particles [30,31]. Thus, the effects of different flow regimes, inflow gas temperature, ambient conditions, thermal conductivity of the solid phase, dispersion coefficient and porosity variation over radius on the gasification process and a thorough comparison between 2D and 3D domains must be assessed.

2. Model formulation

2.1. Problem description

The model was formulated to simulate the filtration combustion of air over a carbon char packed-bed under a laminar flow regime. Fig. 1 depicts the geometry and mesh used for all simulations. A 2D continuum numerical approach was used within the porous media model module available in the commercial software Ansys® Fluent. The domain is divided into three cell zones, i.e., inflow, porous (blue color in Fig. 1(b)), and outflow zones, where the latter is added to obtain a steady-state and stable converged solution. The continuum model is the physical representation of a volume where solid and gaseous phases coexist. This space can be characterized by a porosity (ϵ), which elementally indicates the void fraction, i.e., the available volume where

Table 1

Porous zone characteristics and dimensions.

Parameter	Symbol	Value
Bed porosity	ϵ	0.5063
Inertial resistance	C_2	2378.113 m ⁻¹
Inverse of permeability	$\frac{1}{\kappa}$	8.985 · 10 ⁶ m ⁻²
Number of particles	N_p	85
Particle diameter	d_p	5.6 mm
Tube diameter	d_t	21.2 mm
Tube length	L_t	45 mm

Table 2

Reaction mechanism, reaction rates and kinetics parameters for char gasification.

Equation	Reaction	R_i	A	E_a [J/mol]	n_r
R1	$2C_{(s)} + O_2 \rightarrow 2CO$	$k_{r,O_2} C_{O_2}, s$	$1.5 \cdot 10^5$	$1.494 \cdot 10^5$	0
R2	$C_{(s)} + CO_2 \rightarrow 2CO$	$k_{r,CO_2} C_{CO_2}, s$	4.605	$1.751 \cdot 10^5$	1
R3	$C_{(s)} + H_2O \rightarrow CO + H_2$	$k_{r,H_2O} C_{H_2O}, s$	11.25	$1.751 \cdot 10^5$	1
R4	$CO + 0.5O_2 \xrightarrow{H_2O} CO_2$	$k_{r,CO} C_{CO} C_{H_2O}^{0.5} C_{O_2}^{0.25}$	$1.25 \cdot 10^{11}$	$1.67 \cdot 10^5$	0
R5	$CO + H_2O \rightarrow CO_2 + H_2$	$k_{r,CO} C_{CO} C_{H_2O}$	$2.74 \cdot 10^6$	$8.36 \cdot 10^4$	0
R6	$CO_2 + H_2 \rightarrow CO + H_2O$	$k_{r,CO_2} C_{CO_2} C_{H_2}$	$1.0 \cdot 10^8$	$6.28 \cdot 10^4$	0

the oxidant mixture flows through. For further calculations, the bulk bed porosity was calculated as follows [32]:

$$\epsilon = 1 - \frac{N_p \cdot \frac{1}{6} \pi d_p^3}{\frac{1}{4} \pi d_t^2 L_t} \quad (1)$$

where ϵ is the bed bulk porosity, N_p is the number of particles, d_p is diameter of particles, d_t and L_t are the diameter and the length of the tube, respectively. Parameters used in this study for the porous zone of the reactor are given in Table 1. The diameter of particles inside the packed-bed (d_p) was set to 5.6 mm, following experimental data found in the literature for porous media combustion [12,13] and HFC technologies [14,33]. The inflow gas consisted of several mass fractions of O_2 ($Y_{O_2,in} = 0.05$ and 0.11), 0.001 mass fraction of H_2O (nearly dry air), the rest is nitrogen ($Y_{N_2} = 1 - \sum_i Y_i$). The inflow gas temperature (T_{in}) was variable in the range of 900–1500 K, taking the temperature requirements (~ 1100 K) for achieving thermal conversion of coal in the presence of an oxidant mixture/gasifying agent [34–37]. For the particular application of producing H_2 /syngas from carbon-based feedstocks using HFC, a temperature range of 900–1800 K can be achieved at different scenarios (Equivalence ratio, filtration velocity) due to the enhanced heat transfer mechanisms within the reactor [38,39]. The particle Reynolds number (Re_{in}) was defined using Eq. (2), considering the superficial velocity ($\vec{u}_{in}$), which was calculated using Re_{in} , particle size (d_p), density (ρ_{in}) and viscosity (μ_{in}). Re_{in} of 50, 75, and 100 were tested in this study.

$$Re_{in} = \frac{\rho_{in} \vec{u}_{in} d_p}{\mu_{in}} \quad (2)$$

The chemical kinetics mechanism uses a semi-global scheme consisting of six reactions, with three heterogeneous occurring in the porous phase (Reactions R1 to R3) and three homogeneous in the gaseous phase (Reactions R4 to R6). Specifically, R2 refers to the Boudouard reaction, R4 characterizes the oxidation of carbon monoxide (CO) with water serving as a catalyst species, and R5 and R6 represent the forward and backward water-gas shift (WGS) reactions, respectively. Reaction mechanism, kinetic parameters and reaction rates of both heterogeneous and homogeneous reactions were based on the extended Arrhenius law, mathematically described by Eq. (3), and listed in Table 2 [21,40,41]:

$$k = AT^{n_r} e^{-\frac{E_a}{RT}} \quad (3)$$

where k is the rate constant for reaction, A is the pre-exponential factor, E_a is the activation energy, R is the universal gas constant, T is the temperature, and n_r is the temperature exponent.

2.2. Governing equations of particle-resolved approach

Eqs. (4) to (7) describe the mathematical conservation of mass, momentum, species, and energy for the PRS model [28,42–44], taking into consideration the following assumptions:

- 1 The volatilization of the particles is not included, due to the steady-state feature of the model.
- 2 Thermal equilibrium is assumed for solid and gas phases.
- 3 The gaseous phase behaves as an incompressible flow with $p = 10^5$ Pa.
- 4 The buoyancy effect is neglected.

Taking into account all assumptions made, the final system of equations has the following form [28]:

Conservation equation for mass transport:

$$\nabla \cdot (\rho \vec{u}) = 0 \quad (4)$$

Conservation equation for momentum transport:

$$\nabla \cdot (\rho \vec{u} \vec{u}) = -\nabla p + \nabla \cdot \tau \quad (5)$$

Conservation equation for species transport:

$$\nabla \cdot (\rho \vec{u} Y_i) = \nabla \cdot (\rho D_i \nabla Y_i) + R_i \quad (6)$$

Conservation equation for energy transport:

$$\nabla \cdot (\rho \vec{u} h) = \nabla \cdot (\lambda \nabla T) - \sum_i \frac{h_i^0}{M_i} R_i \quad (7)$$

In the context of the PRS approach, Stefan flow was considered, i.e., when the velocity is non-zero at chemically reacting surfaces, which characterizes the net mass flux between surface and gas [45,46]. Below, we outline the mathematical representation of this phenomenon and provide the boundary conditions for heat and mass transfer.

$$\rho_{wall} D_i \frac{\partial Y_{i, wall}}{\partial n} - \dot{m}_C'' Y_{i, wall} = M_i \hat{R}_{i,s} \quad (8)$$

$$\dot{m}_C'' = \sum_{i=1}^{N_R} M_i \hat{R}_{i,s} \quad (9)$$

$$\mathbf{n} \cdot \lambda \nabla T|_{gas} = \sum_{j=1}^N M_j \hat{R}_{j,s} h_j^0 + \epsilon_s \sigma (T_{surf}^4 - T_{in}^4) \quad (10)$$

$$\mathbf{n} \cdot \vec{u} = \frac{\dot{m}_C''}{\rho} \quad (11)$$

where $\mathbf{n}$ is the normal vector to the wall, ϵ_s is the emissivity ($\epsilon_s \approx 1.0$ for solid carbon), σ is the Stefan–Boltzmann constant, $\hat{R}_{i,s}$ is the production rate of species i due to surface chemical reactions, $\dot{m}_C''$ is the net mass flux between surface and gas, indexes *wall* and *surf* refer to gas side at the wall and surface of the particles, respectively.

2.3. Governing equations of porous media approach

The 2D axisymmetric steady-state model for the conservation equations (Eqs. (12) to (15)) describing PMM is based on several assumptions, as outlined in the literature [47,48]. These assumptions include the composition of the porous zone with spherical carbon char particles, considering fuel particles as a continuously porous medium with initially isotropic properties, excluding drying and devolatilization effects, assuming thermal equilibrium within the char particle between solid and gas phases, setting the ambient gas phase as isobaric with $p = 10^5$ Pa, and assuming laminar gas flow with an axisymmetric flow field.

Taking into account all assumptions made, the final system of equations has the following form [47,48]:

Conservation equation for mass transport:

$$\nabla \cdot (\epsilon \rho_g \vec{u}) = S_m \quad (12)$$

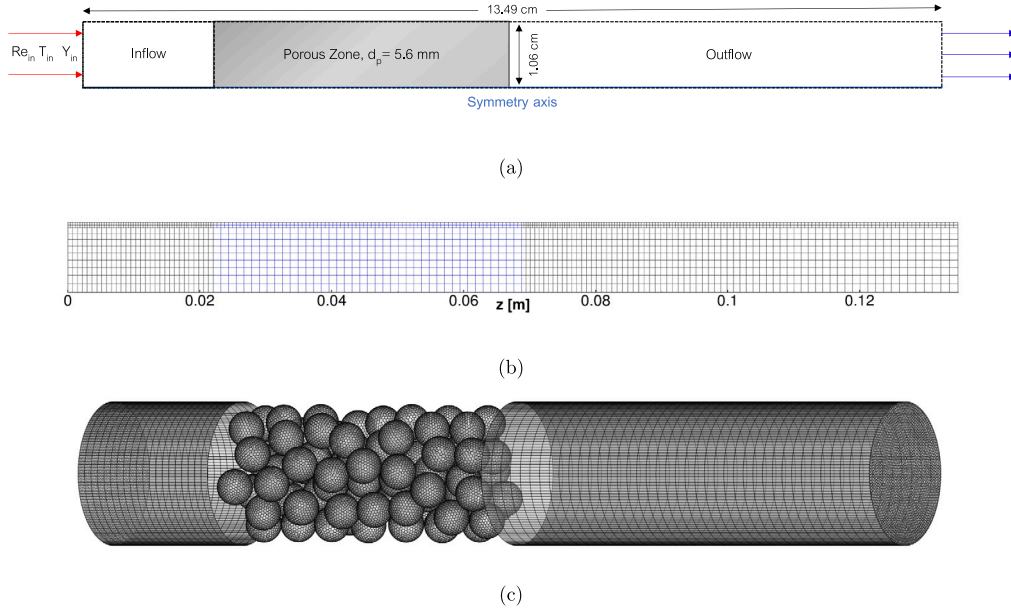


Fig. 1. Schematics of computational domain (a) and mesh of fixed-bed reactor (b) using PMM approach. Mesh of 3D PRS approach assessed in [28] (c). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Conservation equation for momentum transport:

$$\nabla \cdot (\epsilon \rho \vec{u} \vec{u}) = -\epsilon \nabla p + \nabla \cdot (\epsilon \tau) \quad (13)$$

Conservation equation for species transport:

$$\nabla \cdot (\epsilon \rho \vec{u} Y_i) = \nabla \cdot (\epsilon \rho D_{eff} \nabla Y_i) + R_i \quad (14)$$

Conservation equation for energy transport:

$$\nabla \cdot (\epsilon \rho \vec{u} h) = \nabla \cdot (\lambda_{eff} \nabla T) - \sum_i \epsilon \frac{h_i^0}{M_i} R_i \quad (15)$$

where $\tau = \mu (\nabla \vec{u} + \nabla \vec{u}^T)$ is the stress tensor. Species i is any reacting participant involved in the heterogeneous combustion process, i.e., H_2O , O_2 , H_2 , CO , CO_2 , D_{eff} is the mass diffusion coefficient, λ is the thermal conductivity, h_i is enthalpy, h^0 is the enthalpy of formation, M_i is molecular weight, Y_i is mass fraction, R_i is the net production rate, and $\hat{R}_{i,r}$ is the Arrhenius molar rate of creation/destruction for species i , respectively. The last two terms are described below, where $v'_{i,r}$ and $v''_{i,r}$ are the stoichiometric coefficients for reactants and products in the reaction r , $\eta'_{j,r}$ and $\eta''_{j,r}$ are the forward and backward rate exponents in the reaction r for each chemical species j .

$$R_i = M_i \sum_r \hat{R}_{i,r} \quad (16)$$

$$\hat{R}_{i,r} = (v'_{i,r} - v''_{i,r}) \left[k_r \prod_{j=1}^N C_{j,r}^{(\eta'_{j,r} - \eta''_{j,r})} \right] \quad (17)$$

The source term incorporated in the continuity equation may be expressed as:

$$S_m = S''' \left(\rho \frac{M_C}{M_{O_2}} k_{R1} Y_{O_2} + \rho \frac{2M_C}{M_{O_2}} k_{R2} Y_{O_2} + \rho \frac{M_C}{M_{CO_2}} k_{R3} Y_{CO_2} \right) \quad (18)$$

The density of the gaseous mixture was calculated as stated by Eq. (19). Viscosity and thermal conductivity were calculated using the kinetic theory [49].

$$\rho = \frac{P}{RT \sum_i \frac{Y_i}{M_i}} \quad (19)$$

Regarding the PMM approach, the porosity ($\epsilon = 0.5063$) impacts on the laminar flow through a packed bed. The Blake–Kozeny equation may be used for calculating pressure drop, as described in Eq. (20) [50].

The inverse of permeability ($\frac{1}{k}$) and inertial loss coefficient of flow (C_2) in the porous zone are described by Eqs. (21) and (22), respectively.

$$\frac{|\Delta p|}{L} = \frac{150 \mu (1 - \epsilon)^2}{D_p^2 \epsilon^3} v_\infty \quad (20)$$

$$\frac{1}{\kappa} = \frac{D_p^2}{150 (1 - \epsilon)^2} \epsilon^3 \quad (21)$$

$$C_2 = \frac{3.5 (1 - \epsilon)}{D_p \epsilon^3} \quad (22)$$

The effective diffusivity ($D_{m,i}^{eff}$) was calculated using Eq. (23):

$$D_{m,i}^{eff} = \frac{\epsilon}{\tau} \frac{1 - x_i}{\sum \frac{x_i}{D_{ij}}} \quad (23)$$

where ϵ is porosity; τ is tortuosity; x_i is molar fraction of i th species; D_{ij} is binary diffusivity for $i - j$ th pair. D_{ij} is calculated from:

$$D_{ij} = \frac{0.01013 T^{1.75}}{p_i M^{0.5} (V_i^{1/3} + V_j^{1/3})^2} \quad (24)$$

where $V_{i,j}$ — diffusion volume for i, j th species [51], i.e., $V_{CO_2} = 26.9$, $V_{CO} = 18.9$, $V_{H_2} = 7.07$, $V_{N_2} = 17.9$, $V_{O_2} = 16.6$, $V_{H_2O} = 12.7$. The thermal conductivity (λ_{eff}) inside the porous particle was calculated using the porosity of the fixed-bed:

$$\lambda_{eff} = \epsilon \lambda \quad (25)$$

2.4. Boundary conditions

At the inlet of the fixed-bed reactor, the following Dirichlet boundary conditions are considered for inflow gas velocity, species mass fractions, and temperature, respectively:

$$\vec{u} = \vec{u}_{in}, \quad Y_i = Y_{i,in}, \quad T = T_{in} \quad (26)$$

where $\vec{u}_{in} = f(Re_{in})$, $T_{in} = 900 - 1500$ K, and $Y_{i,in}$ varies depending on the oxidative atmosphere ($Y_{O_2,in} = 0.05$ and 0.11). At the outlet of the fixed-bed reactor, the so-called outflow boundary conditions with an overall mass balance correction was implemented [49]. Since there is an important interaction between gaseous and solid phases on the surface of the particles, mass and energy balance at the interface are

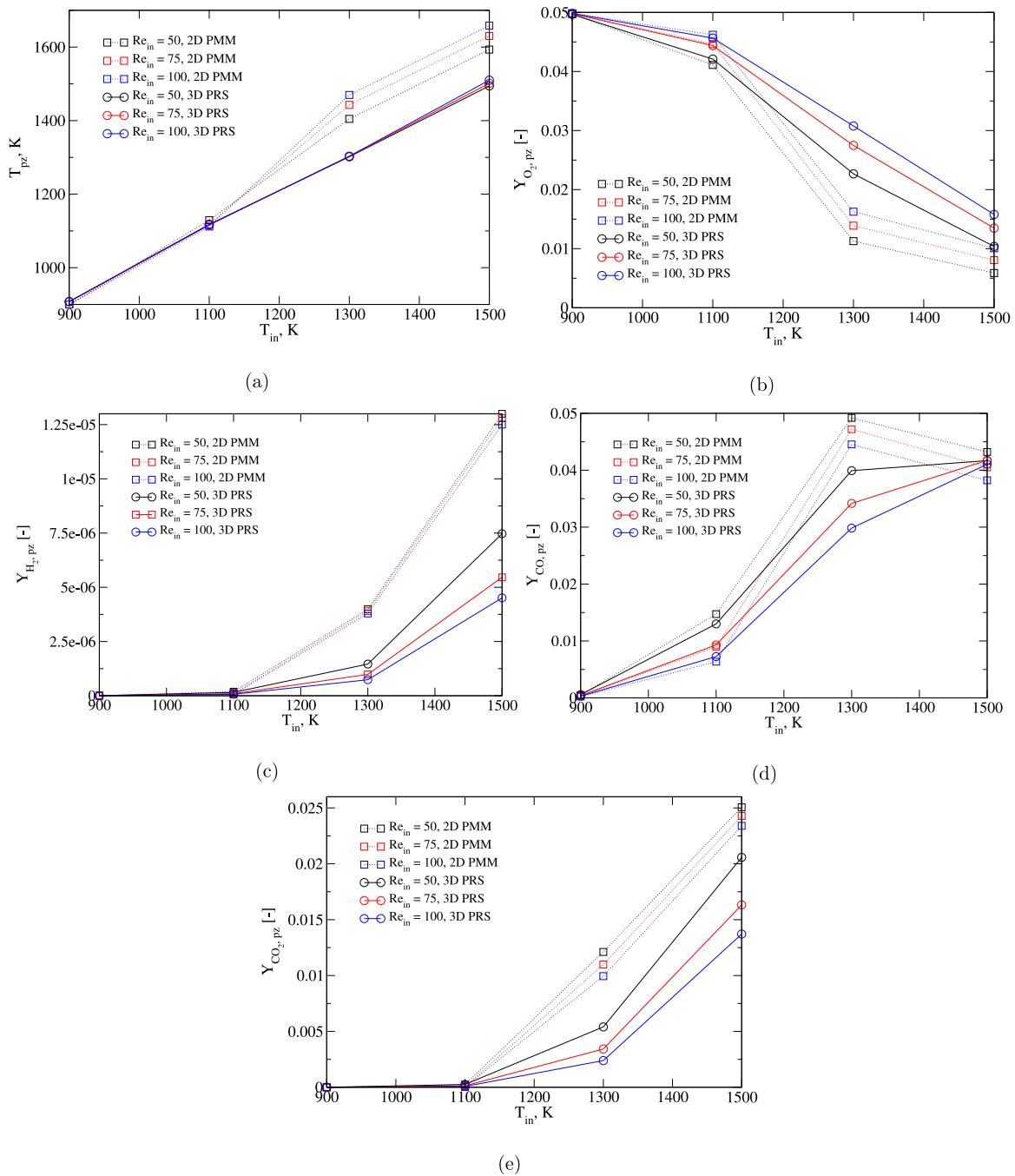


Fig. 2. Volume-averaged T_{pz} (a), Y_{O_2} (b), Y_{H_2} (c), Y_{CO} (d), and Y_{CO_2} (e) inside porous zone for $Y_{O_2,in} = 0.05$ and $Re_{in} = 50, 75$ and 100 .

affected by the heterogeneous chemical reactions (R1 to R3), due to the production and destruction rates of gaseous species caused by these reactions along with convective and diffusive mass fluxes of the gas phase species at the surface of particles [52]. A crucial parameter for heterogeneous chemical reactions is the specific surface area of the porous zone, which was calculated taking into account the number of particles and their occupied volume inside the unit, being $S''' = 528.99 \text{ m}^{-1}$ the value of specific surface area used for both PMM and PRS numerical approaches.

Regarding the PMM approach, at the symmetry axis ($r = 0$) as well as at the upper side of the domain, the following Neumann boundary conditions must be specified for pressure, velocity, concentration, and temperature:

$$\frac{\partial p}{\partial r} = 0, \quad \frac{\partial \vec{u}}{\partial r} = 0, \quad \frac{\partial Y_i}{\partial r} = 0, \quad \frac{\partial T}{\partial r} = 0 \quad (27)$$

2.5. Numerics and validation

Numerical software Ansys® Fluent 2022R2 [49] was utilized to solve Eqs. (12) to (14). The solution method for solving the governing equations by coupling pressure and velocity was the semi-implicit method for pressure-linked equations (SIMPLE) algorithm [53]. The scheme of the computational domain and grid structure are schematically depicted in Figs. 1(a) and 1(b), respectively. The porous zone can be identified in Fig. 1(b) for the blue mesh. The number of cells was 1960, to ensure uniform porosity along the bed zone. It should be noted that the grid independence simulations for 3D results have been reported in Arriagada et al. [28]. In terms of 2D grid study, we must emphasize that in porous media models the grid size is limited by the size of particles. In particular, if a grid size in 2D PMM simulation is significantly less than particle diameter, then in this case, volume fraction

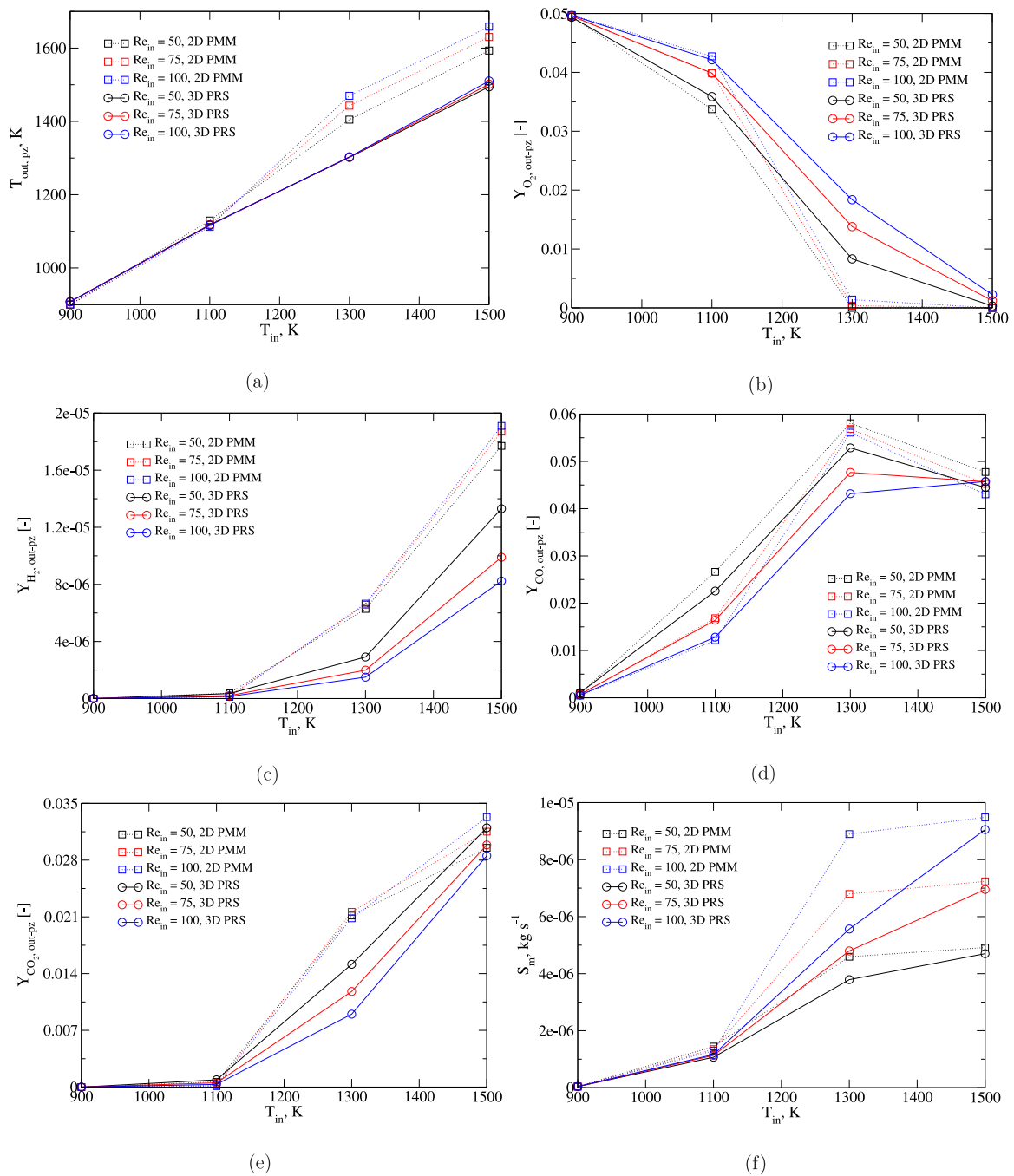


Fig. 3. Area-weighted averaged T_{out-pz} (a), Y_{O_2} (b), Y_{H_2} (c), Y_{CO} (d), and Y_{CO_2} (e) at outlet of the porous zone. S_m (f) for $Y_{O_2,in} = 0.05$ and $Re_{in} = 50, 75$ and 100 .

radial and axial profiles must be included into the simulations because of the volume, which is further discussed in Section 3.6. Specific surface area (S''') was used for considering wall-surface reactions in the porous zone. Radiation was considered using the P-1 model [54] since it plays a crucial role as a heat transfer mechanism inside the porous zone of the reactor. Convective terms, species, and energy were discretized using an upwind second-order scheme. The under-relaxation factors were set at 0.5 for momentum, energy 0.8, and species 0.5, except for carbon dioxide (0.7), and oxygen (0.3), as the kinetic scheme involved in the process has a critical effect on these species' conservation. The iterations were stopped when the residual for all equations was less than 10^{-6} , which required approximately 10^5 iterations until convergence. The software was validated against both experimental data [19,42,55] and analytical solutions [18], considering a two-film model [56], and

the same kinetic scheme implemented in the present work. Functions for effective diffusivity and thermal conductivity in the porous zone were implemented into the software through a user-defined-function (UDF), where the DEFINE macros of DEFINE_DIFFUSIVITY and DEFINE_PROPERTY were used respectively.

3. Results and discussion

3.1. Effect of heterogeneous chemical reactions and Reynolds number

Reynolds numbers based on particle size (Re_{in}) of 50, 75, and 100 were numerically studied in the reactor. From an elemental perspective, as Re_{in} increases, a higher inlet velocity ($\bar{u}_{in}$) is observed. When comparing 3D CFD particle-resolved simulations against the porous media

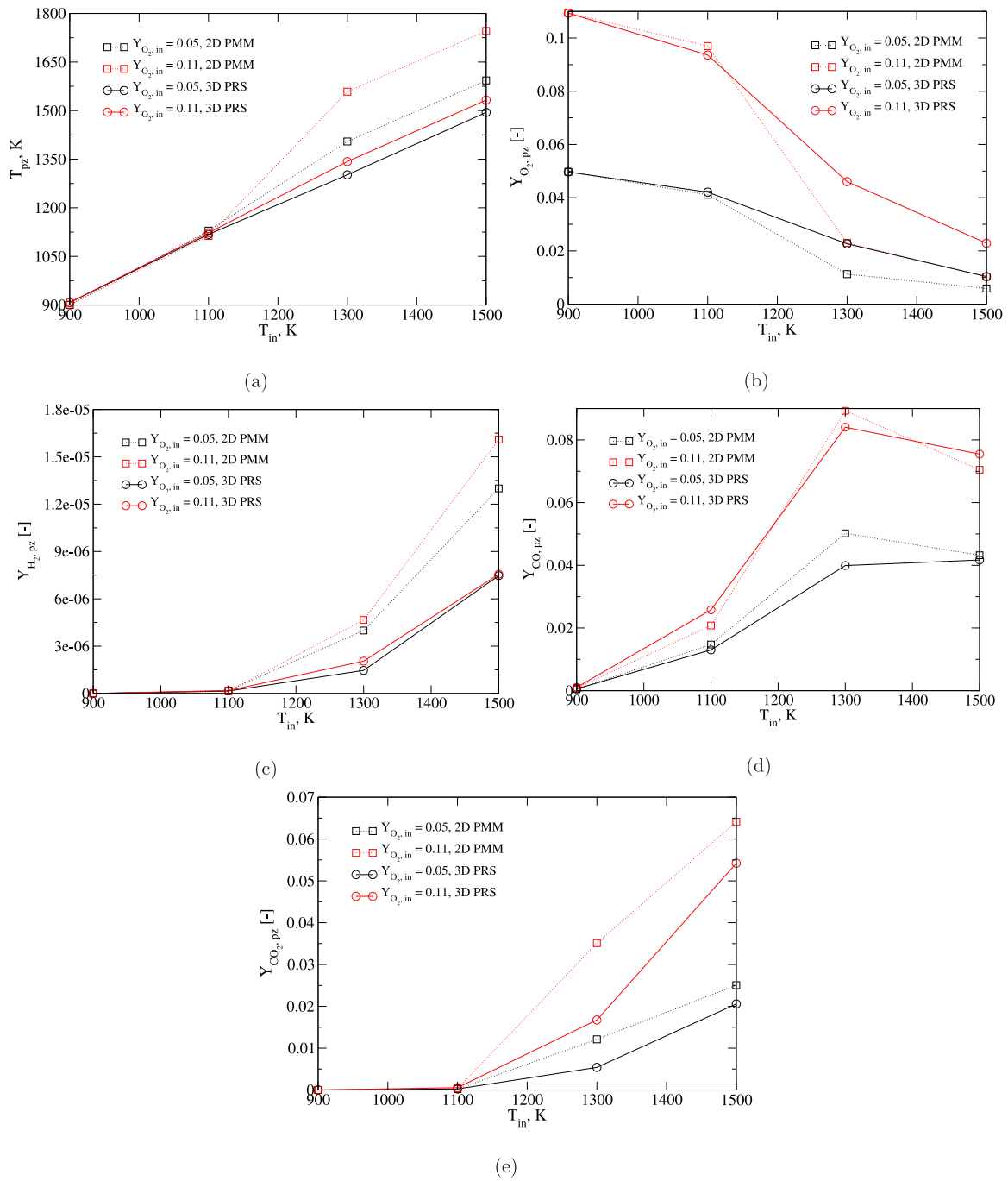


Fig. 4. Volume-averaged T_{pz} (a), Y_{O_2} (b), Y_{H_2} (c), Y_{CO} (d), and Y_{CO_2} (e) inside porous zone for $Re_{in} = 50$, and $Y_{O_2, in} = 0.05$ and 0.11 .

model in terms of the effect of the heterogeneous chemical reactions in the species concentration, Fig. 2 depicts the volume-averaged temperature (Fig. 2(a)) and mass fractions of oxygen (Fig. 2(b)), hydrogen (Fig. 2(c)), carbon monoxide (Fig. 2(d)), and carbon dioxide (Fig. 2(e)). It can be observed how Re_{in} affects the profiles, particularly observing for oxygen how the concentration decreases with inflow gas temperature, reaching almost full consumption in the porous zone at high inlet temperature. The highest deviation observed for the temperature inside the porous zone (T_{pz}) is presented for $Re_{in} = 100$, with 12.8%. The effect of R3 and R5 could be observed in Fig. 2(c), where the hydrogen mass fraction increases with the inflow gas temperature. Here the order of magnitude of the mass fraction of hydrogen (Y_{H_2}) increases from an average of $5 \cdot 10^{-6}$ to $1.2 \cdot 10^{-5}$. Moreover, the effect of R4 and R5 could be seen in Fig. 2(d), where between 1300 K and 1500

K, the mass fraction of CO does not increase with a positive slope as compared with hydrogen or carbon dioxide. It could be highlighted that carbon dioxide is quite accurate in all simulations, with a deviation below 12.3% in all cases when comparing PRS and PMM. From 1100 K and higher temperatures, the slope of the curve representing the mass fraction of CO_2 increases remarkably, which could be attributed to R2 and R4, dominant chemical reactions in the semi-global kinetic scheme used for this work. In general terms, the 2D axisymmetric approach is in agreement with results obtained in the literature for 3D PRS modeling [28], specially for low Reynolds number and low inflow gas temperature, where the profiles follow a relatively linear behavior as the temperature increases. The exception is $T_{in} = 1300$ K, where the highest deviation between the 2D PMM and 3D PRS approaches occurs, which is consistent with the literature when evaluating the

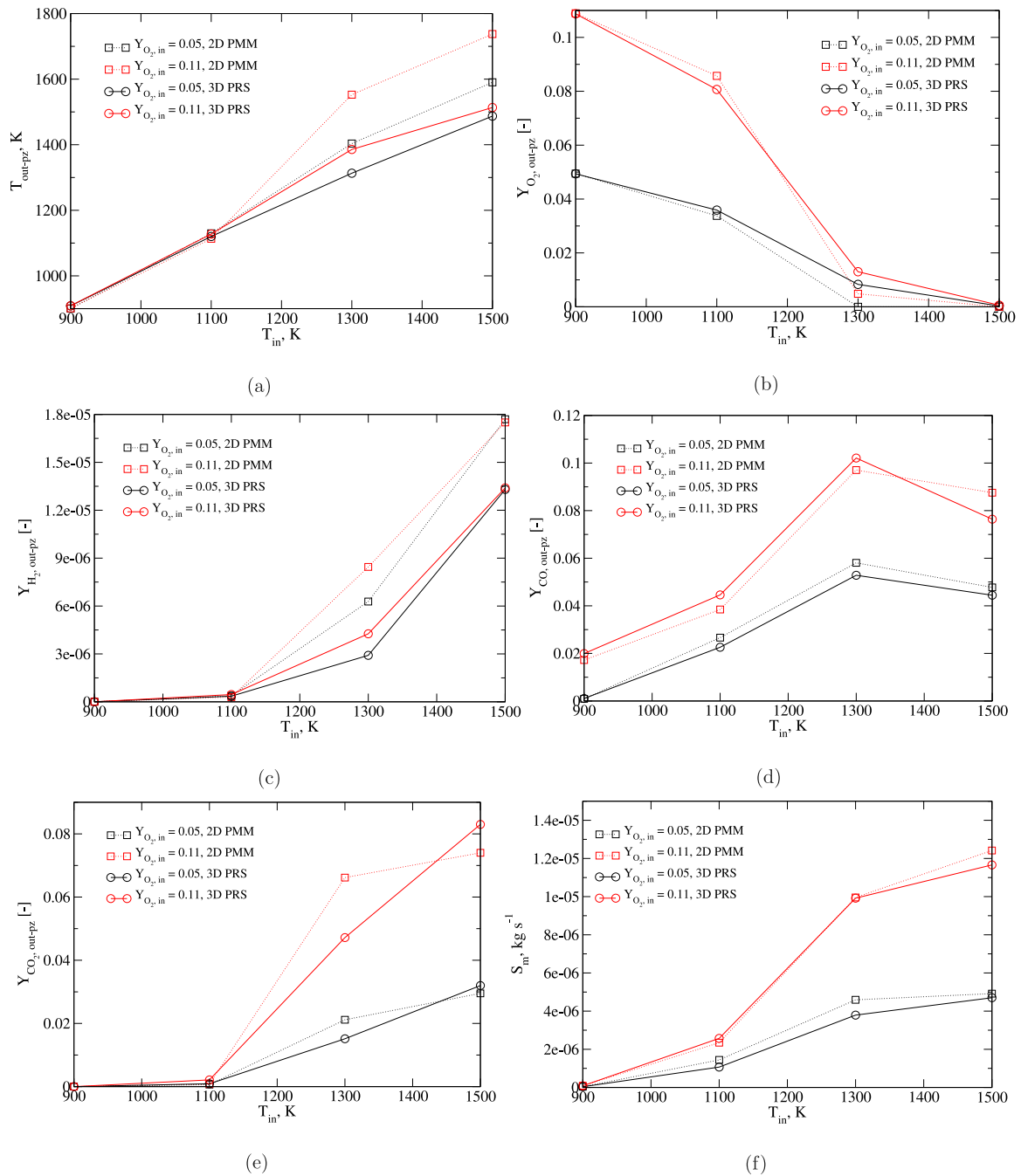


Fig. 5. Area-weighted averaged T_{out-pz} (a), Y_{O_2} (b), Y_{H_2} (c), Y_{CO} (d), and Y_{CO_2} (e) at outlet of the porous zone. S_m (f) for $Re_{in} = 50$, and $Y_{O_2,in} = 0.05$ and 0.11 .

effect of inflow gas temperature on the heterogeneous combustion of carbon [57]. Furthermore, the effect of Re_{in} can be clearly observed, where the higher the inflow velocity, the higher the deviation exists between the two numerical approaches under study.

This tendency can also be observed in Fig. 3, where the intention was to evaluate the effect of the continuum model against a particle-resolved simulation approach, without considering homogeneous reactions taking place after the porous zone. To evaluate the accuracy of the porous media model, the following results of the aforementioned figure are calculated at the outlet of the bed zone ($z = 6.88$ cm). The area-weighted averaged temperature (T_{out-pz}) and concentration of main species involved in the kinetic scheme show the same behavior as for the volume-averaged values, with higher maximum concentration of hydrogen (See Fig. 3(c)), carbon monoxide (See Fig. 3(d)), and carbon dioxide (See Fig. 3(e)) when compared these

two results. Again, deviation between PRS and PMM increases with Re_{in} , where a particular control point might be found at $T_{in} = 1300$ K. At this temperature, most of the available O_2 is consumed, CO_2 presents deviation smaller than 16.7% for all cases, where exact values are 7.7%, 5.5%, and 16.7% for $Re_{in} = 50, 75$, and 100 , respectively. It can also be observed a minimum deviation of 8.1% at H_2 $T_{in} = 1100$ K, which is a negligible value since the order of magnitude of Y_{H_2} is 10^{-7} . The effects of the Boudouard reaction (R2) and the oxidation of CO (R4) are remarkable for these species since this is evaluated at the outlet of the porous zone. The effect of the PMM approach was evaluated at the outlet of the porous zone since after the packed-bed, changes in flow velocity, formation of velocity gradients, generation of eddies or recirculation zones may affect the comparison with a 3D PRS approach. Regarding the mass flow, there must be a difference between outflow and inflow mass. There has been a minimal difference between these

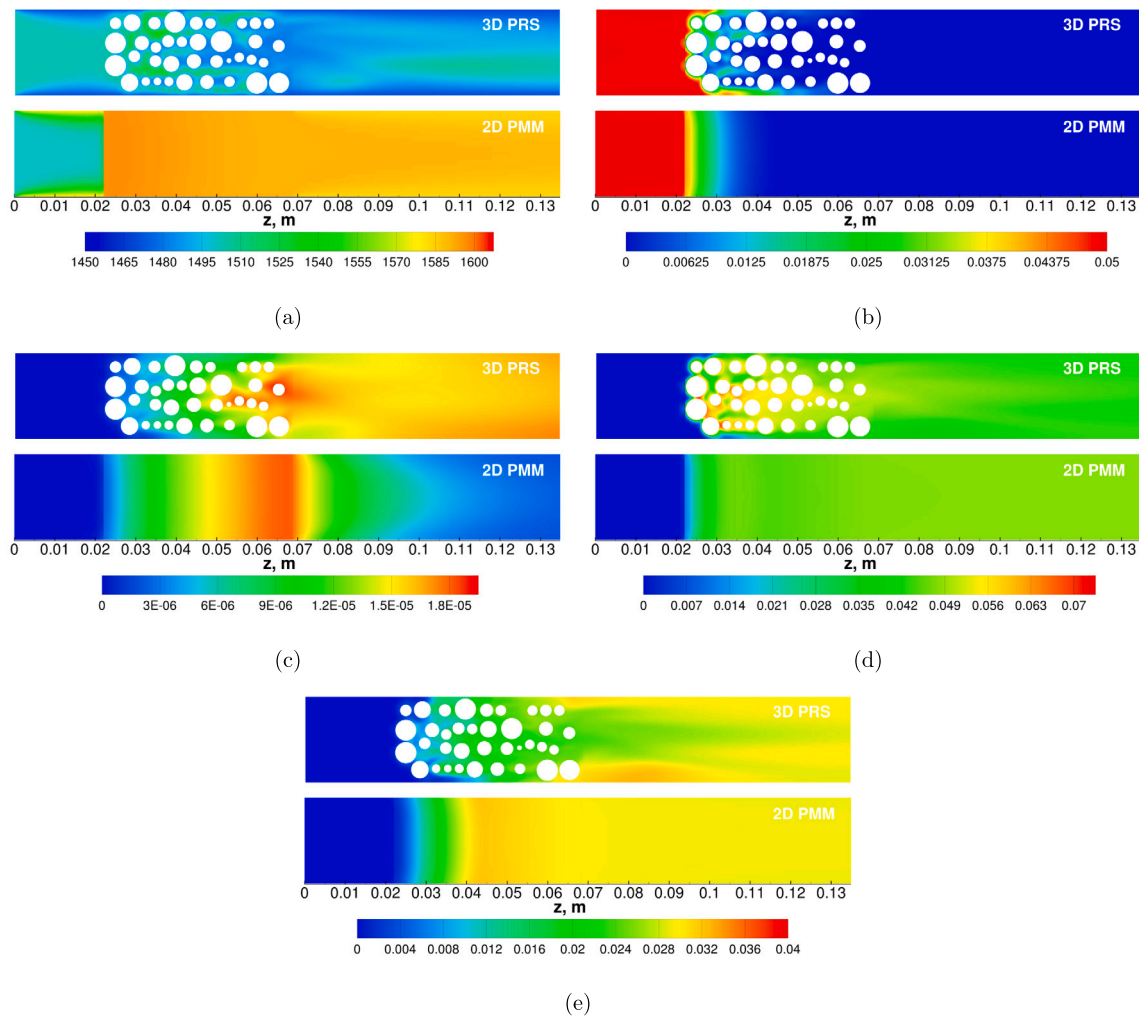


Fig. 6. Temperature (a), Y_{O_2} (b), Y_{H_2} (c), Y_{CO} (d), and Y_{CO_2} (e) inside fixed-bed reactor for $Re_{in} = 50$, $Y_{O_2,in} = 0.05$, and $T_{in} = 1500$ K.

two approaches. Fig. 3(f) shows the highest difference for $Re_{in} = 100$, $T_{in} = 1300$ K, with $3.36 \cdot 10^{-6}$ kg s $^{-1}$. The tendency of the results is similar to those reported previously, where the higher the Reynolds number, the more deviation with results when comparing 2D PMM and 3D PRS.

3.2. Effect of oxidative atmosphere

Two different gasifying conditions were evaluated during this research. Thermochemical processes using an ambient mass fraction of oxygen $Y_{O_2} = 0.05$, and 0.11 were used to assess the behavior of the domain, and performance of the heterogeneous combustion of the fixed-bed reactor. Fig. 4 shows the volume-averaged temperature (Fig. 4(a)) and mass fractions of oxygen (Fig. 4(b)), hydrogen (Fig. 4(c)), carbon monoxide (Fig. 4(d)), and carbon dioxide (Fig. 4(e)). It can be observed a good agreement between both 2D axisymmetric PMM and PRS numerical approaches for all the presented profiles. In particular, and as discussed in the previous section, the highest deviation could be observed for an inflow gas temperature of 1300 K, while the higher this variable, the higher deviation between 2D and 3D models might be seen for all curves. Most probably, the effect of a higher volume-averaged temperature inside the porous zone leads to both a decrease of the oxygen concentration (See Fig. 4(b)), and greater production of carbon dioxide (See Fig. 4(e)). Deviation on average does not get higher than 10% for temperature, whereas for the lowest inflow temperature, the deviation is below 1%, showing a good fit of curves.

In particular, minimum deviations for $T_{in} = 900$ K are 0.85%, and 0.76% for $Y_{O_2,in} = 0.05$ and 0.11, respectively. On the other hand, maximum deviations are observed at $T_{in} = 1300$ K, with 7.9% and 16.1% for the aforementioned case. Also, oxygen and carbon monoxide (See Fig. 4(d)) show the best fit in terms of tendency, achieving a remarkable accuracy for low inflow gas temperatures (900–1100 K). In particular, minimum deviations occur at $T_{in} = 1500$ K, with 3.6% and 6.6% for $Y_{O_2,in} = 0.05$ and 0.11, respectively. As evaluated in the previous section with Re_{in} as the crucial parameter to assess, Fig. 5 depicts the effect of the PMM approach against PRS at the outlet of the porous zone ($z = 6.88$ cm), avoiding any constrain due to the abrupt transition between the porous zone and the outlet zone of the unit, comparing $Y_{O_2,in} = 0.05$ and 0.11 at a constant value of $Re_{in} = 50$. Figs. 5(b) and 5(d) show excellent agreement between both PMM and PRS, in particular for oxygen leading to the entire conversion of oxygen at high inflow gas temperatures, and the highest production of CO for inflow gas temperature of 1300 K for all cases under study.

3.3. Effect of homogeneous chemical reactions

As observed in the previous subsections of this article, the porous media model is specially different when compared to a 3D CFD particle-resolved simulation in terms of the fluid dynamics inside the fixed-bed. Fig. 6 shows the temperature and species concentration along the reactor under both 3D PRS (top view) and 2D PMM (bottom view) at $Re_{in} = 50$, $Y_{O_2,in} = 0.05$, and $T_{in} = 1500$ K. It could be observed how the

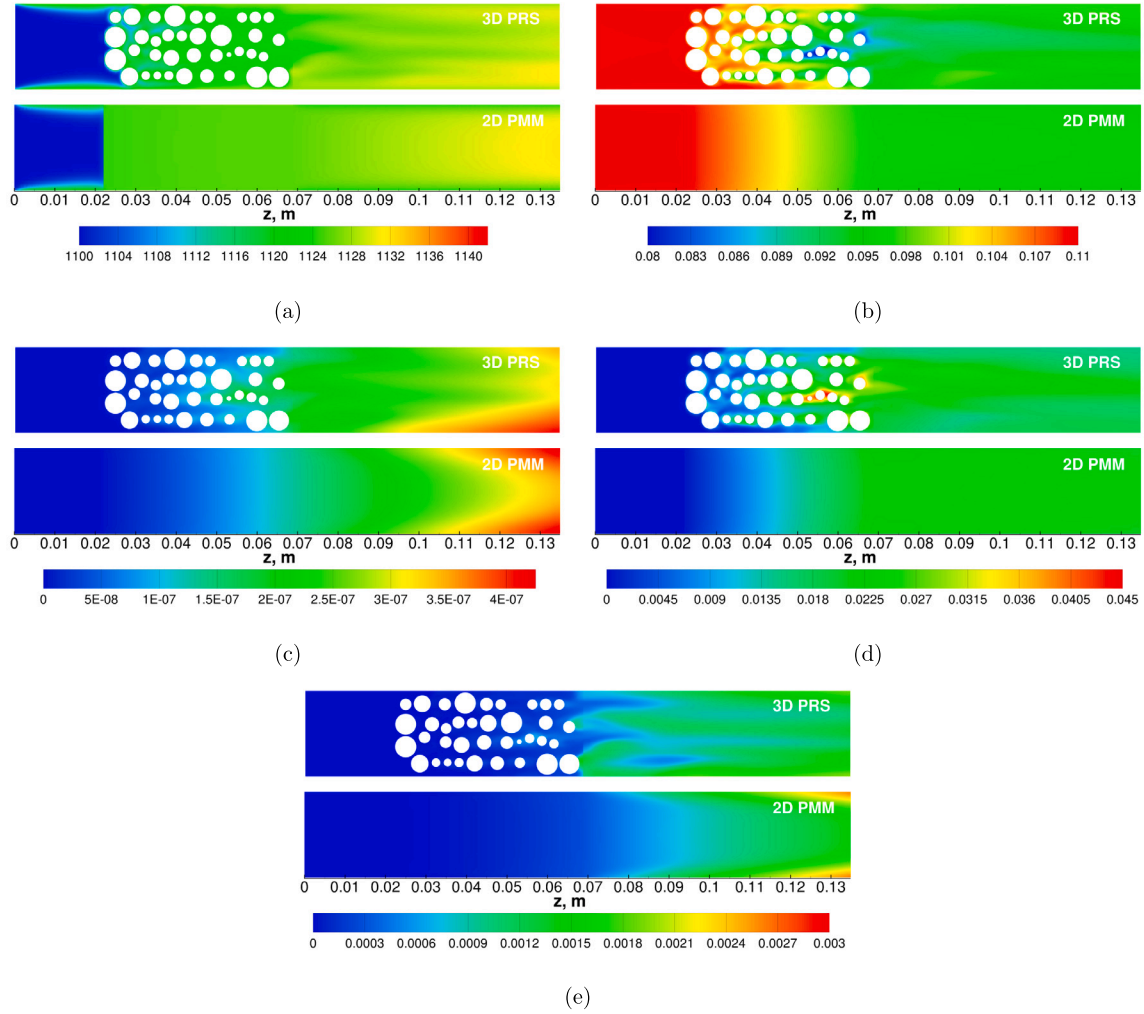


Fig. 7. Temperature (a), Y_{O_2} (b), Y_{H_2} (c), Y_{CO} (d), and Y_{CO_2} (e) inside fixed-bed reactor for $Re_{in} = 100$, $Y_{O_2,in} = 0.11$, and $T_{in} = 1100$ K.

effect of homogeneous chemical reactions after the porous zone plays a crucial role. In particular, and for hydrogen (See Fig. 6(c)), hydrogen is consumed in the 2D PMM approach after the porous zone. Moreover, the profiles are relatively accurate for oxygen and carbon monoxide but do not show a good agreement for temperature, hydrogen, and carbon dioxide. On the other hand, for a lower inflow gas temperature, such as $T_{in} = 1100$ K, Fig. 7 depicts again temperature, and main species mass fractions along the reactor using both numerical approaches, observing an excellent agreement between 3D PRS and 2D PMM, not only in the magnitude but also in the position along the reactor, which represents the effect of both heterogeneous and homogeneous chemical reactions.

3.4. Effect of thermal conductivity of solid phase

The heterogeneous combustion of carbon char particles inside a fixed-bed is a continuous interaction between the solid and gas phases. The thermal conductivity of the solid phase is an important parameter to assess. Fig. 8 shows the effect of the variation of thermal conductivity of the solid phase inside the porous zone (λ_s) on the volume-averaged temperature and species concentration. This test was done considering $Y_{O_2,in} = 0.05$ and $Re_{in} = 75$, and adding a third plot, considering $\lambda_s = 0.02$ W/m K. In general, it can be observed how the lower λ_s value used for simulations, better accuracy of the 2D PMM model is achieved, due to the surface-based reaction model studied by Arriagada et al. [28], where no mesh inside solid phase existed. In particular, for the critical inlet temperature of $T_{in} = 1300$ K, when the thermal conductivity of the

solid phase was treated near air conditions (hollow spheres), the curves fit better decreasing the gap between 3D PRS and 2D PMM. Results are consistent with those reported in the literature, where λ_s plays a crucial role in the heat transfer inside packed-bed reactors [32].

3.5. Effect of dispersion coefficient

The dispersion coefficient might play a relevant role in reaction phenomena inside a packed-bed reactor, depending on their geometrical features, such as D_t/d_p ratio, L_t/d_p ratio, Peclet number (Pe_m), among others. This parameter has been widely studied in the literature with several approaches depending on the operational conditions and dimensions of a packed-bed reactor [58–61]. To assess the effect of the dispersion coefficient in the performance of the 2D PMM approach, the longitudinal dispersion coefficient D_L , described by Eq. (28), was implemented using the macros DEFINE_DIFFUSIVITY in a user-defined-function [61–63]:

$$D_L = 0.7D_{ij} + ur_p \quad (28)$$

The effect of D_L is depicted in Fig. 9, where the volume-averaged temperature (Fig. 9(a)), Y_{O_2} (Fig. 9(b)), Y_{H_2} (Fig. 9(c)), Y_{CO} (Fig. 9(d)), and Y_{CO_2} (Fig. 9(e)) inside the porous zone are presented. It can be observed negligible deviation between 2D PMM (constant ϵ , no consideration of D_L) and 2D PMM (constant ϵ , including D_L) is negligible in the range of 900–1100 K. Deviation gets considerable for higher inflow gas temperatures ($T_{in} = 1300$ –1500 K) for all figures. Considering the

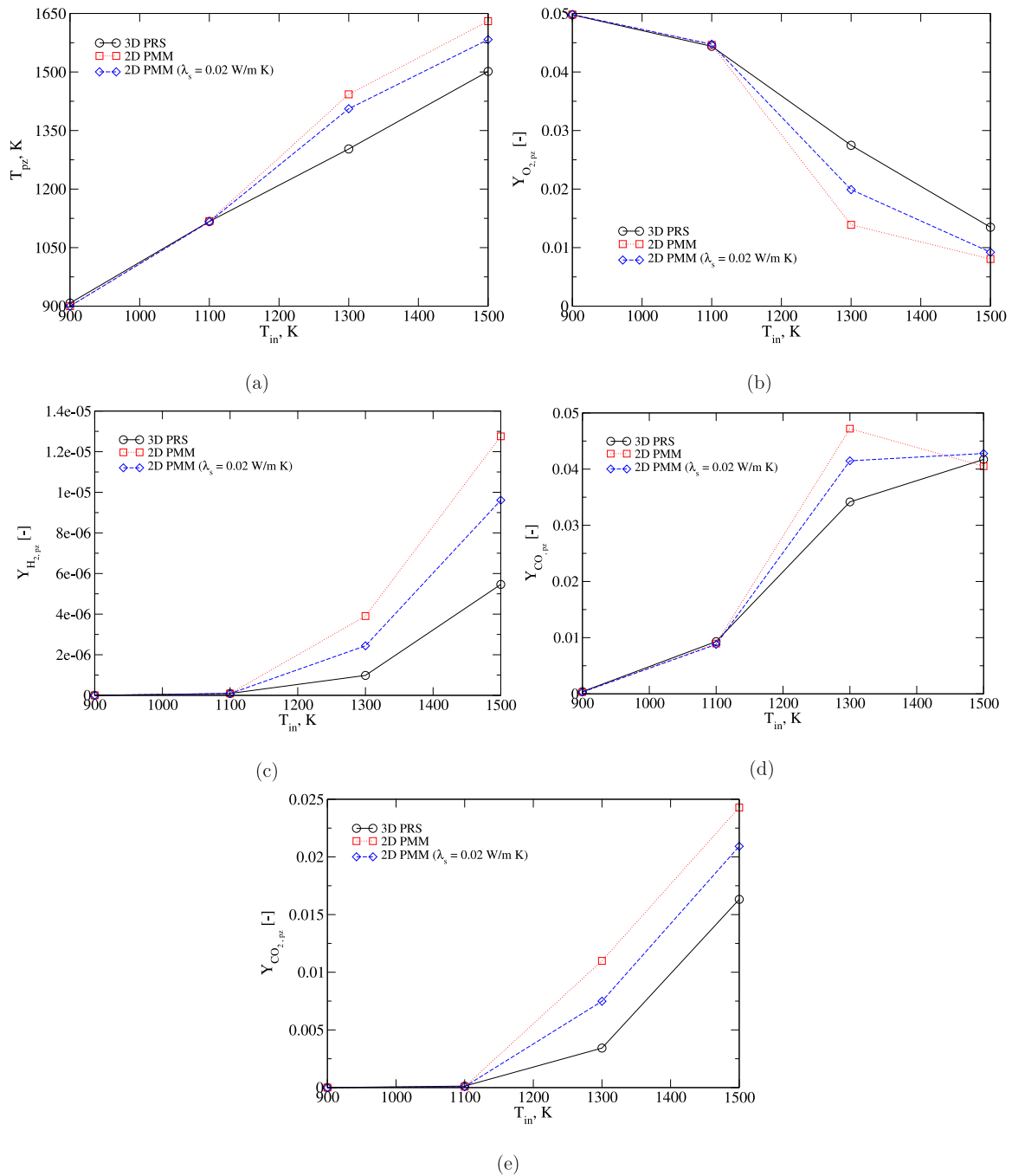


Fig. 8. Effect of thermal conductivity of solid (λ_s) on the volume-averaged T_{pz} (a), Y_{O_2} (b), Y_{H_2} (c), Y_{CO} (d), and Y_{CO_2} (e) inside porous zone for $Y_{O_2,in} = 0.05$ and $Re_{in} = 75$.

axial dispersion coefficient is significant for the partial oxidation of a chemically reacting fixed-bed, thus should be considered when using a 2D PMM numerical approach.

3.6. Effect of porosity variation

The porosity plays a crucial role when working with continuum models such as the porous media model (PMM) as it depends on the D_t/d_p ratio [64]. Moreover, the porosity inside a packed-bed reactor varies with the radius ($\epsilon(r)$) depending as well on the D_t/d_p ratio, showing different mathematical correlations [65–69]. A derivation of a particular mathematical representation of $\epsilon(r)$ considering the specific features of the reactor ($D_t/d_p = 3.78$) under study is described

in Eq. (29) [66]. In this case, a refined mesh was necessary to perform simulations since the profile in the radial direction behaves in a non-structured way.

$$\epsilon(r) = \begin{cases} 2.14z^2 - 2.53z + 1; & z \leq 0.637 \\ \epsilon_b + 0.29\exp(-0.6z)[\cos(2.3\pi(z - 0.16))] & \\ + 0.15\exp(-0.9z); & z > 0.637 \end{cases} \quad (29)$$

where $z = (R_t - r)/d_p$, r is the radius, d_p is particle diameter, $R_t = 10.6$ mm is the tube radius, and ϵ_b is bulk porosity, which in this case takes the value of 0.37. The function of $\epsilon(r)$ was implemented into the software through a user-defined-function (UDF), where the DEFINE_PROFILE macros was used. Since there is a continuous change in the value of porosity over radius of the reactor, a refined mesh was used, refining the whole domain, specially at the wall, obtaining a

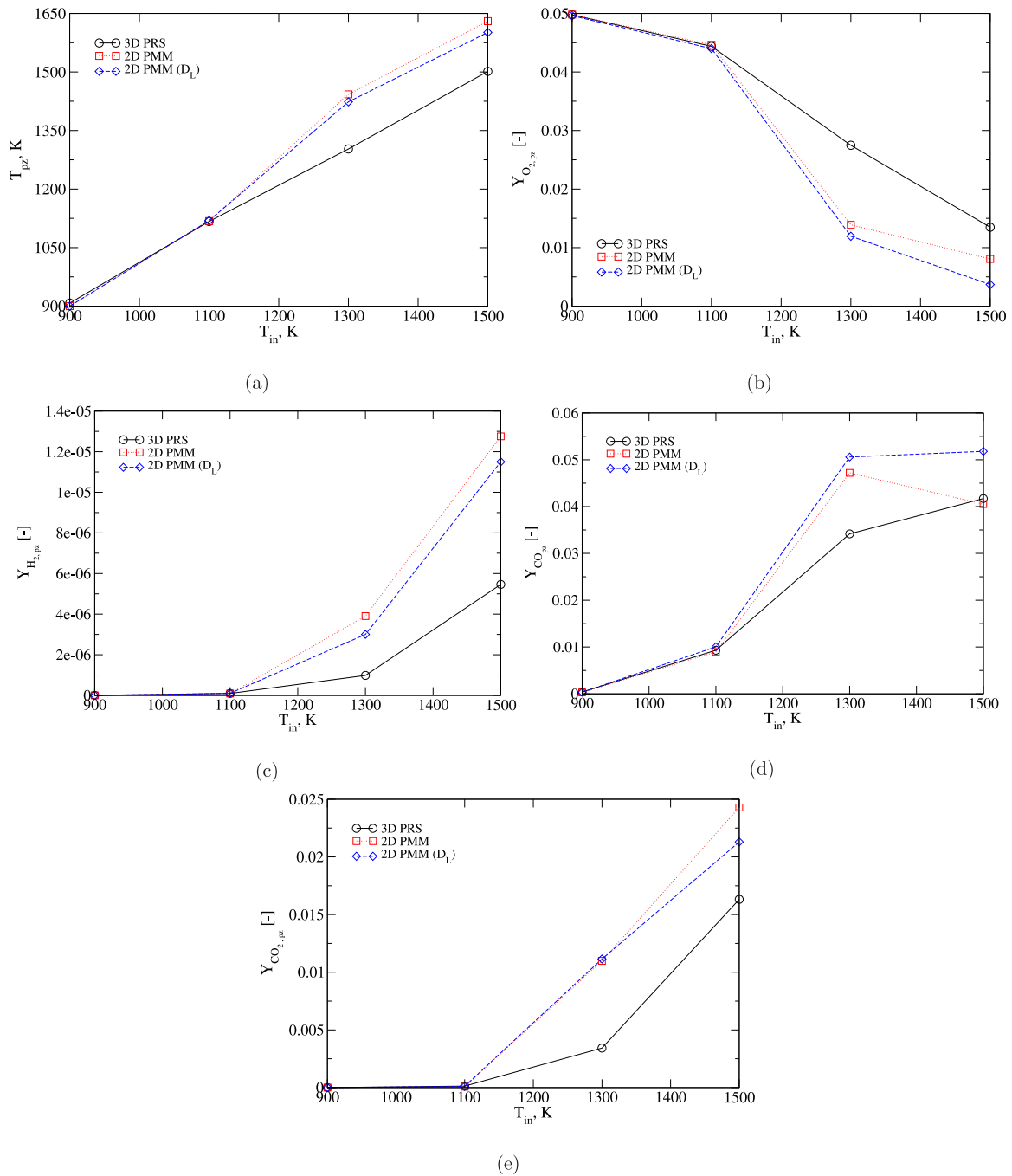


Fig. 9. Effect of dispersion coefficient (D_L) on the volume-averaged T_{pe} (a), Y_{O_2} (b), Y_{H_2} (c), Y_{CO} (d), and Y_{CO_2} (e) inside porous zone for $Y_{O_{2, in}} = 0.05$ and $Re_{in} = 75$.

18 500 cell mesh. Fig. 10 shows the effect of incorporating a function of $\epsilon(r)$ in the numerical simulations. Fig. 10(a) shows a comparison of $\epsilon(r)$ using the approach of De Klerk [66] and Mueller [69]. The latter was not later considered due to the sharpness of the function at a near value of $z = 1$. For figures representing volume-averaged temperature (Fig. 10(b)), Y_{O_2} (Fig. 10(c)), Y_{H_2} (Fig. 10(d)), Y_{CO} (Fig. 10(e)), and Y_{CO_2} (Fig. 10(f)), an important deviation comparing with using constant porosity, i.e., $\epsilon = 0.5063$, can be observed. The consideration of a function of porosity for narrow tubes, in this case, $D_t/d_p = 3.78$, plays a crucial role in how the gasification process occurs inside the fixed-bed.

Adding a function for $\epsilon(r)$ improved the accuracy of the 2D PMM fitting to the 3D PRS case, even for an inlet temperature of 1300 and 1500 K. Fig. 11 depicts contour plots of temperature (Fig. 11(a)), Y_{O_2} (Fig. 11(b)), Y_{H_2} (Fig. 11(c)), Y_{CO} (Fig. 11(d)), and Y_{CO_2} (Fig. 11(e))

showing the effect of considering a function for porosity on the most critical inflow gas temperature discussed in this study ($T_{in} = 1300$ K). The difference between 2D PMM and 2D PMM $\epsilon(r)$ is evident since including porosity as a function of radius leads to the models to be in good agreement. Moreover, the 2D PMM model including $\epsilon(r)$ shows consistency with chemical kinetics, showing consumption of oxygen, and generation of H_2 , CO , and CO_2 at different rates. At high temperatures, the 2D PMM captures the gasification performance accurately inside the porous zone, deviating in some of the species (H_2 and CO) due to the effect of the oxidation of carbon monoxide and water-gas-shift reaction. Results highlight the relevance of geometrical features of the unit. If the diameter of the tube increases accordingly with the number of particles, a reduction of the effect of wall channeling would be expected, both decreasing the temperature of the solid particles

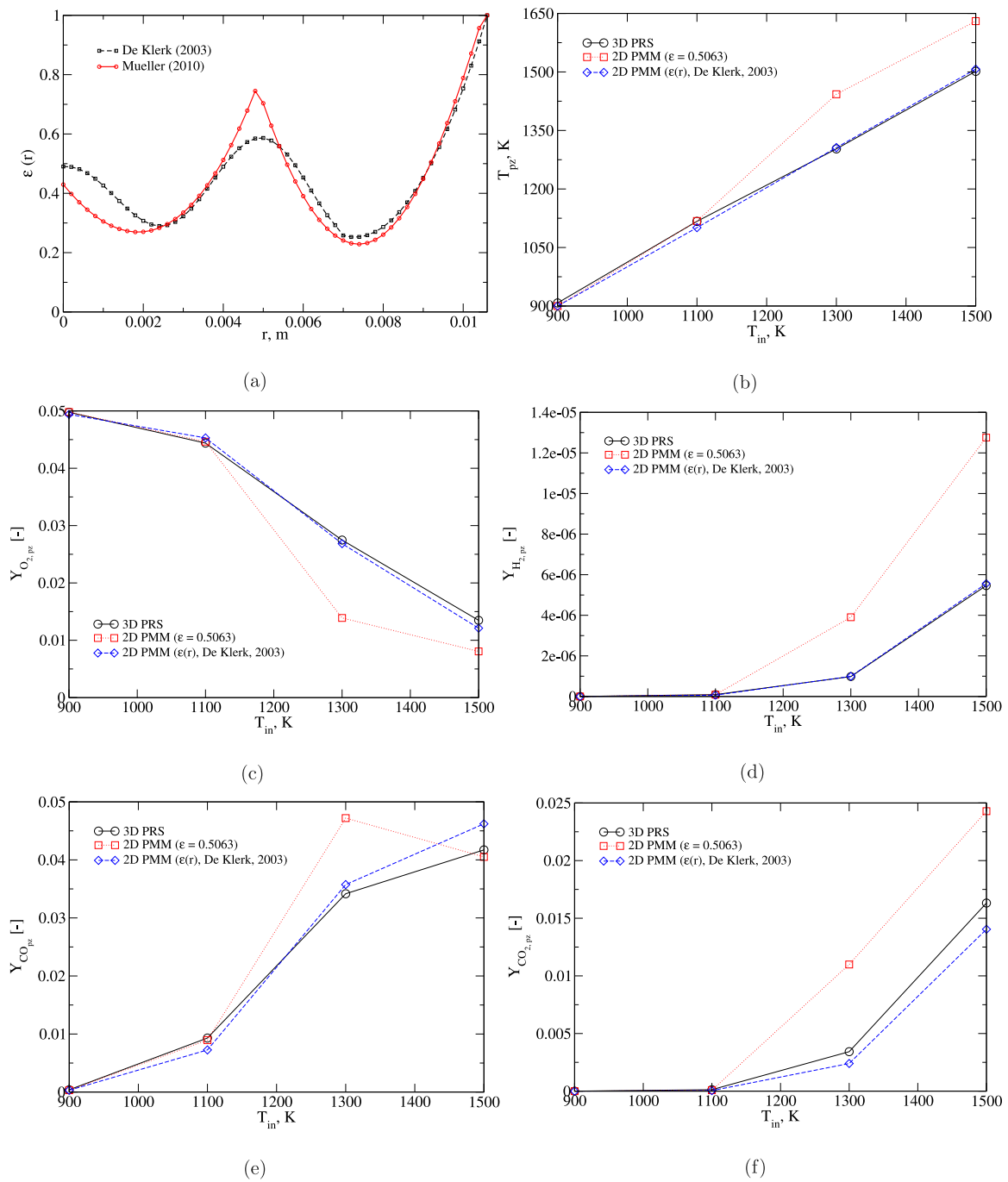


Fig. 10. Porosity function over reactor radius ($\epsilon(r)$) (a), effect of $\epsilon(r)$ on volume-averaged T_{pz} (b), Y_{O_2} (c), Y_{H_2} (d), Y_{CO} (e), and Y_{CO_2} (f) inside porous zone for $Y_{O_2,in} = 0.05$ and $Re_{in} = 75$.

along Z axis and increasing Y_{CO} faster. Moreover, for inert porous media reactors, where normally alumina spheres are used for enhancing heat transfer mechanism inside the fixed-bed, decreasing the particle size will lead to higher temperatures inside the unit.

4. Concluding remarks

In this work, the filtration combustion of carbon char particles under a laminar flow regime was investigated using a 2D axisymmetric porous media CFD-based model. The PMM approach was verified against 3D CFD surface-based particle-resolved simulations (PRS) for the partial oxidation of a chemically reacting fixed-bed, assessing critical parameters of flow regime (Re_{in}), oxidative atmosphere ($Y_{O_2,in}$),

inflow gas temperature (T_{in}), thermal conductivity of solid phase (λ_s), variable porosity ($\epsilon(r)$), and axial dispersion coefficient (D_L).

Results showed the significant role of the inflow gas temperature T_{in} in the gasification process. In particular, an important activation of chemical reactions is observed for high inlet temperatures (1300 K), showing considerable deviations when compared with the 3D PRS model. Minimum deviations were observed for $Y_{O_2,in} = 0.05$, being 3.6% for CO, 5.5% for CO_2 and 0.76% for the temperature inside the porous zone at different operating conditions. The effect of the axial dispersion coefficient (D_L) was also assessed, showing a considerable effect on the performance of the partial oxidation at high inflow gas temperatures ($T_{in} = 1300$ –1500 K), when comparing 2D PMM approaches, thus, becoming a significant parameter to consider

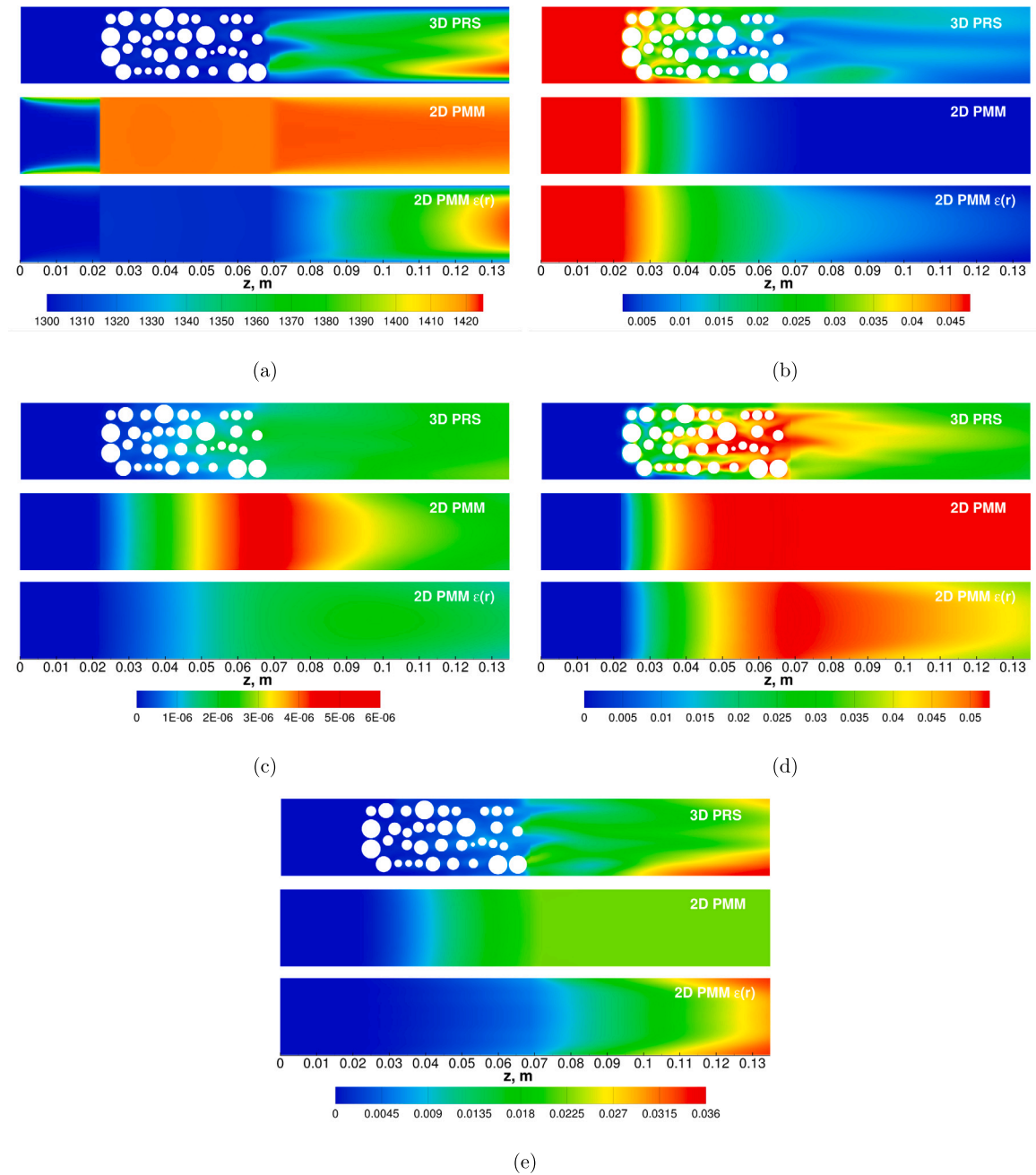


Fig. 11. Contour plots of chemical species mass fractions (Y_i) predicted numerically using 3D PRS, 2D PMM, and 2D PMM for non-constant volume fraction of solid ($\epsilon = f(r)$), see (Fig. 10(a)) for $Re_{in} = 75$, $Y_{O_2,in} = 0.05$, and $T_{in} = 1300$ K: T_{pz} (a), Y_{O_2} (b), Y_{H_2} (c), Y_{CO} (d), and Y_{CO_2} (e). Top plot shows results of 3D PRS, down plot depicts 2D PMM with $\epsilon = f(r)$ and the middle plot shows 2D PMM with constant ϵ .

when simulating this phenomenon, where it is also crucial to consider geometrical features of the reactor, such as D_t/d_p and L_t/d_p . Furthermore, the effect of the thermal conductivity of the solid phase (λ_s) was discussed, fitting with good accuracy to the 3D PRS baseline model, incorporating a negligible value for λ_s , assuming hollow spheres (surface-based model).

Finally, the use of a uniform porosity value along the fixed-bed was validated by simulating under the same conditions a variable porosity over radius $\epsilon(r)$. Variation of the porosity showed significant differences with those using constant porosity due to the narrow D_t/d_p ratio used for this study, achieving good agreement with the 3D PRS model. In addition, results highlighted the relevance of geometrical features of the reactor, distinguishing effects of wall channeling and mass fraction

dynamics of carbon monoxide and on heat transfer mechanism inside inert porous media reactors.

There is an important space to explore regarding numerical approaches for thermochemical conversion processes. Further studies are necessary to assess intrinsic chemical reactions (macro and micro-pore-level numerical studies, tracking of porosity and particle radius) for a single particle and for fixed-bed reactor problems, variation of the converted fuel and gasifying agent, geometrical features of the unit (D_t/d_p), and hybridization of the porous zone.

CRediT authorship contribution statement

Andrés Arriagada Romero: Writing – review & editing, Writing – original draft, Visualization, Software, Investigation, Formal analysis.

Mario Toledo: Writing – review & editing, Supervision, Investigation, Funding acquisition, Formal analysis. **Robert E. Hayes:** Writing – review & editing, Visualization, Supervision, Software, Funding acquisition, Formal analysis, Conceptualization. **Dmitry Pashchenko:** Software, Formal analysis. **Petr A. Nikrityuk:** Writing – review & editing, Visualization, Supervision, Software, Resources, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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